Muhammed Raseel P

Flutter Developer | Mobile App Engineer

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github.com/Raseelp raseelportfolio.vercel.app

ABOUT ME

Flutter Developer with **1.5 years of professional experience** and nearly **3 years of hands-on experience working with Flutter and native Android development in Kotlin**, with strong experience using **GetX** and integrating **REST APIs**, along with testing and debugging using **Postman**. I have **strong native Android experience** building performance-critical, low-level modules including on-device machine learning pipelines and **custom Flutter–Kotlin bridge architectures**, that go well beyond what the Flutter framework handles on its own. I have collaborated closely with **.NET-based backend** systems at work and explore backend development using **Go and MongoDB** through personal projects. Experienced in using **Firebase services** in production to **improve app stability, performance, and reliability**. I have experience releasing and maintaining production-ready cross-platform applications on both the **Play Store and App Store.**

SKILL

- **Programming Languages:** Dart, Kotlin, Go, C
- **Frameworks & Technologies:** Flutter, Android (Native), GetX, Provider, Gin, Django, REST APIs
- **Databases:** SQLite, MongoDB, MySQL
- **Firebase:** Crashlytics, Performance Monitoring, App Check, Cloud Messaging (FCM)
- **Tools & Platforms:** Play Store & App Store (Production Releases), Postman, Git, Azure DevOps, GitHub, ffmpeg
- **AI / ML Integration**: ONNX Runtime, PyTorch Mobile Runtime, On-Device Inference, Face Detection & Recognition, CLIP-based Semantic Search

PROFESSIONAL EXPERIENCE

Flutter Developer Full-Time, -- Logiology Solutions 04/2025 – 11/2026 | Calicut, Kerala

Worked as a Flutter Developer in a **product-based company delivering Jewellery ERP solutions**, contributing to production-grade mobile applications used in real business environments.

N Vend – Jewellery Sales & Billing Application for Salesmen

- **Maintained and enhanced N Vend**, a core jewellery sales application, with **Play Store and App Store deployments**.
- **Designed and developed a complete CRM module**, and automated activity and opportunity creation, scheme and receipt workflows, and customer interaction tracking.
- Worked extensively with **complex mathematical and financial calculations**, including multiple taxation models such as **CGST, SGST, IGST, VAT**, ensuring accuracy and compliance.
- Implemented **PDF generation and printing workflows** using the pdf package, with **custom layouts for invoices and proforma documents.**
- Integrated **Firebase Cloud Messaging (FCM)** to enable real-time notifications for business-critical events.
- Built **MRZ parsing** and **debit/credit card data parsing** features to support identity and payment-related workflows.
- Integrated **RFID-based stock-taking,** enabling bulk scanning of tagged inventory through **radio-frequency identification and real-time item tracking.**

N Divo – Jewellery Business Analytics Application for Owners

- Worked on **N Divo**, a jewellery owner-focused business monitoring application, deployed on the **Play Store and App Store** .
- Implemented and enhanced dashboards and reports covering **stock, sales, salesmen performance, profits, assets, liabilities**, and other key business metrics.
- Developed a **branch-level aggregation module**, enabling owners to view **combined data from multiple jewellery branches** in a single, unified view.
- Implemented **biometric authentication** to secure sensitive business and financial data, strengthening application security and access control.
- Integrated **Firebase services** to improve application reliability, monitoring, and real-time updates.

Flutter Developer Intern, -- Regional Technologies 01/2025 – 02/2025 | Calicut, Kerala

- Led a team of 4 developers to create a **real-time emergency vehicle priority system** using Flutter and Django, reducing ambulance response times by **30%.**
- Collaborated on **geofencing notifications, SOS requests, and a medical chatbot**, enhancing user engagement by **20%.**

PERSONAL PROJECTS

Local Semantic Search & Face Recognition Engine 12/2025 – Present
(Flutter, Kotlin, CLIP Vision/Text Encoders, ONNX Runtime,PyTorch Mobile Runtime, SQLite)

- Developed an **offline semantic search and face recognition system** that lets users **search local images and videos using natural language queries** and automatically **identifies and groups the people** in them, matching on the actual visual content inside media files, **entirely on-device with no cloud processing.**
- Designed and implemented a **multimodal semantic retrieval pipeline** using **CLIP-based vision and text encoders**, enabling natural-language and image-based search across local media collections.
- Developed an embedding generation workflow that transforms images, video frames, and text queries into **shared high-dimensional vector representations** for cross-modal similarity search.
- Implemented vector similarity ranking using **dot-product / cosine similarity** operations over **locally indexed embeddings** to retrieve semantically relevant results without cloud inference.
- Designed and implemented an **on-device face recognition pipeline (SCRFD-2.5G detection, ArcFace recognition)** using **ONNX Runtime**, architected so the **detector and recognition model are model-agnostic** and swappable without code changes.
- Built an **incremental clustering algorithm** that groups **recognised faces into people in real time** as photos are indexed, using per-person prototype embeddings and exemplar-based comparison, with automatic merge suggestions and user-correctable grouping (merge, split, undo).
- Extended the **recognition pipeline to video, sampling frames, linking face detections across frames** into tracks, and batching recognition per track to identify people in video content at a fraction of the naive per-frame inference cost.
- Built a **self-contained model lifecycle system: background downloads with checksum verification, resumable transfers, progress reporting, and versioned on-device storage**, for both the search and face-recognition models.
- Built **a self-correcting tool** that automatically detects when two different people were mistakenly grouped together as one, and splits them apart accurately (using a **multi-seed 2-means clustering algorithm**) — tested and verified at **100% accuracy** across **40 real test cases** before releasing it as a **one-tap fix** for users.
- Solved a **resource-contention problem** where the app's photo indexing, video thumbnail loading, and face scanning all competed for the **phone's limited hardware video decoders** at the same time, causing silent failures — **built a shared coordination system** so these processes take turns instead of colliding.
- Designed an on-device **SQLite store** for faces and people supporting incremental writes from a background worker, safe concurrent access during scanning, and schema migrations across app versions.
- Developed the Flutter application layer and Flutter–Kotlin bridge architecture using **MethodChannels/EventChannels** to support real-time indexing progress, media retrieval, and hybrid native/Flutter interaction.

iPhone-to-Android Video Compatibility Converter, (Kotlin, Media3 Transformer, FFmpeg) 09/2026

- Built a **native Android app** that fixes iPhone videos that fail to play on Android, converting them into a **universally supported format** by combining several different video processing engines to **maximize compatibility** across devices.
- Designed a **multi-step fallback system** that automatically retries a conversion using a different strategy whenever the first one fails, since real iPhone HDR/Dolby Vision videos trigger inconsistent, hardware-specific failures depending on the device.
- Integrated **FFmpeg** as a backup video engine for the cases where Android's built-in tools couldn't process a file at all, **along with automatic color correction** so HDR footage still looks accurate after conversion.
- Assembled a custom **FFmpeg processing chain** using color-space and tone-mapping filters **(zscale, tonemap)** for accurate HDR-to-standard color conversion, **hardware-accelerated encoding for speed, and metadata correction** for proper video orientation and instant playback on Android.
- Identified and fixed a hardware limitation where some devices could play 4K video but not create it, by having the app **detect each device's real capabilities** and adjust the output resolution to match automatically.

EDUCATION

Bachelor of Computer Science 2022 – 2025 | Malappuram
College of Applied Science Vazhakkad, Malappuram

CERTIFICATES AND ACHIEVEMENTS

BR Hackathon (Organized by PearAI)
PearAI (Y Combinator W24) Track Winner